Name: ______________________

DECIMALS, (DIVIDING DECIMALS)

Date: ________________

LO: I can divide decimal numbers by whole numbers and by other decimals.

Dividing decimals

To **divide a decimal** by a whole number, use short/long division keeping the **decimal point** in the same position in the answer.

Dividing by a decimal: Multiply both numbers by 10 (or 100) to make the divisor a whole number, then divide as normal.

Key methods

●	8.4 ÷ 4 = 2.1	decimal point stays
●	3.6 ÷ 0.6 = 36 ÷ 6 = 6	multiply both by 10
●	12.5 ÷ 0.5 = 125 ÷ 5 = 25	equivalent division
●	8.4 ÷ 4 = 21	forgot decimal point
●	3.6 ÷ 0.6: only multiply top by 10	must multiply both

Key idea

If dividing by a decimal, multiply both numbers by 10 or 100 to make the divisor a whole number.

3.6 ÷ 0.6 = 36 ÷ 6 = 6

Make the divisor whole

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – decimal by whole number

Calculate 9.6 ÷ 4

Set up short division: 9.6 ÷ 4

9 ÷ 4 = 2 r 1; bring down 16 ÷ 4 = 4

Keep decimal point aligned: 2.4

= 2.4

The decimal point stays in the same place.

Example 2 – by a single-digit

Calculate 15.75 ÷ 5

15 ÷ 5 = 3

7 ÷ 5 = 1 r 2; 25 ÷ 5 = 5

= 3.15

= 3.15

Work digit by digit; carry remainders.

Example 3 – dividing by a decimal

Calculate 4.8 ÷ 0.6

Multiply both by 10: 48 ÷ 6

48 ÷ 6 = 8

= 8

= 8

Equivalent division: both ×10 doesn't change the answer.

Example 4 – dividing by 0.2

Calculate 7.5 ÷ 0.25

Multiply both by 100: 750 ÷ 25

750 ÷ 25 = 30

= 30

= 30

Multiply by 100 because the divisor has 2 decimal places.

YOUR TURN – PRACTICE (deliberate practice)

1. Decimal ÷ whole number
Calculate.

a) 8.4 ÷ 2

b) 9.6 ÷ 4

c) 12.5 ÷ 5

d) 18.6 ÷ 3

2. More decimal ÷ whole number
Keep the decimal point in line.

a) 15.75 ÷ 5

b) 24.8 ÷ 4

c) 3.42 ÷ 6

d) 27.36 ÷ 8

3. Dividing by a decimal (times 10)
Multiply both by 10 first.

a) 4.8 ÷ 0.6

b) 3.5 ÷ 0.7

c) 9.6 ÷ 0.4

d) 7.2 ÷ 0.9

4. Dividing by a decimal (times 100)
Multiply both by 100 first.

a) 7.5 ÷ 0.25

b) 6 ÷ 0.12

c) 10 ÷ 0.05

d) 3.6 ÷ 0.15

5. Mixed division
Calculate each.

a) 14.4 ÷ 6 b) 8.1 ÷ 0.3 c) 20 ÷ 0.4 d) 5.6 ÷ 0.08 e) 0.72 ÷ 0.9

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) 8.4 ÷ 4 = 2.1 ☐

Correct answer: __________

Reason: __________________

b) 3.6 ÷ 0.6 = 0.6 ☐

Correct answer: __________

Reason: __________________

c) Multiply both by 10 to remove decimal divisor ☐

Correct answer: __________

Reason: __________________

d) When dividing by 0.5, the answer is smaller ☐

Correct answer: __________

Reason: __________________

e) 4.8 ÷ 0.6 = 48 ÷ 6 = 8 ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ribbon **7.2 m** long is cut into pieces of **0.45 m** each. How many pieces can be cut? Is there any leftover?

Expression: ______________________________

Simplified: ______________________________

b) A car uses **5.6 litres** of fuel per **100 km**. How far can it travel on **42 litres**? Give your answer to the nearest km.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!